# LLM Alignment: Enhancing Mathematical Reasoning in SmolLM2-360M using Group Relative Policy Optimization (GRPO)

**Bao Gia Khuong**
Bellini College of Artificial Intelligence, Cybersecurity and Computing
University of South Florida
Tampa, FL, USA
giabaokhuong@usf.edu

## Abstract

Small language models (SLMs) frequently struggle with multi-step reasoning tasks, such as mathematical problem-solving. While Supervised Fine-Tuning (SFT) can improve performance by teaching the model how to format a response, it does not explicitly optimize the sequential decision-making process during token generation. In this project, we formulate language model generation as a reinforcement learning problem and apply Group Relative Policy Optimization (GRPO) to SmolLM2-360M on the GSM8K dataset. We conduct a reward shaping study, comparing sparse outcome-based rewards against dense, reasoning-aware reward functions. Our experiments reveal that SFT establishes a strong and difficult-to-beat baseline, achieving the highest format adherence (0.4150) and tying for peak accuracy (4.17%) with GRPO-R2. Among RL variants, no reward function decisively surpassed SFT on accuracy, but R4 successfully reduced average thinking length by 18% compared to SFT (66.9 vs. 81.7 tokens), demonstrating that reward shaping can meaningfully alter generation behavior even when absolute accuracy gains are limited by model capacity.

## 1 Introduction

Reliable multi-step mathematical reasoning remains a challenge for small language models. Supervised Fine-Tuning (SFT) is commonly used to teach structured logic (e.g., using <think> tags), but it fails to explicitly optimize the autoregressive generation process or penalize internal logical inconsistencies. To address this, we frame text generation as a reinforcement learning problem, applying Group Relative Policy Optimization (GRPO) to SmolLM2-360M on the GSM8K benchmark.

Because sparse rewards often fail small models due to credit assignment issues over long generation horizons, we conducted a reward shaping study evaluating four distinct functions ranging from outcome-only checks to dense reasoning-aware signals. By comparing our GRPO-trained models against a robust SFT baseline, we demonstrate that while a 360M parameter model has a apparent accuracy ceiling, applying a reasoning-aware reward successfully matches SFT's peak accuracy while yielding significantly more concise and efficient reasoning chains.

This project makes three contributions:

- Reward shaping comparison for SLMs
- Analysis of GRPO vs SFT at small scale
- Efficiency vs accuracy tradeoff insight

## 2 Background

We treat the language model as a policy (agent) interacting with a static environment. The reinforcement learning framework is formulated as follows:

- **State ($s$):** A mathematical problem prompt accompanied by structured instructions (e.g., requiring the output to include a specific reasoning format).
- **Action ($a$):** The token-by-token autoregressive generation of the solution.
- **Policy ($\pi_\theta$):** The language model (SmolLM2-360M) parameterized by $\theta$.
- **Episode:** One complete generated sequence consisting of the reasoning steps and the final solution.

The core challenge lies in defining a reward signal that evaluates the final outcome while explicitly guiding the model to adopt robust reasoning patterns.

## 3 Method

To optimize the reasoning policy, we implemented GRPO using the TRL library. The model generates $G = 4$ candidate completions per prompt, computes relative rewards within each group, and updates the policy via clipped surrogate loss with a KL penalty to prevent deviation from the reference policy.

We conducted a reward shaping study by evaluating four distinct reward functions to observe their effects on reasoning behavior:

**R1 — Outcome-Based Reward (Baseline)** Assigns a reward of $+1$ strictly if the final parsed answer is correct, and 0 otherwise:

$$R_1 = \begin{cases} 1 & \text{if } \hat{a} = a^* \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

**R2 — Format + Outcome Reward** Grants partial rewards for adhering to the required XML structure and answer correctness:

$$R_2 = 0.2 \cdot f_{\text{think}} + 0.2 \cdot f_{\text{answer}} + 0.6 \cdot f_{\text{correct}} \tag{2}$$

where $f_{\text{think}} = 1$ if <think>...</think> is present, $f_{\text{answer}} = 1$ if <answer>...</answer> is present, and $f_{\text{correct}} = 1$ if $\hat{a} = a^*$, else 0.

**R3 — Reasoning-Aware Reward (Main Contribution)** Provides partial rewards for the presence of intermediate arithmetic steps within the <think> block, while still weighting final correctness highest:

$$R_3 = \min(0.3,\ n_{\text{steps}} \times 0.05) + 0.7 \cdot f_{\text{correct}} \tag{3}$$

where $n_{\text{steps}}$ is the count of arithmetic expressions (e.g., `5 + 9`, `12 * 3`) detected via regex within the reasoning chain, and $f_{\text{correct}} = 1$ if $\hat{a} = a^*$, else 0.

**R4 — Efficiency-Regularized Reward** Introduces a length penalty for completions exceeding 150 characters, encouraging more concise solutions:

$$R_4 = \max\left(0,\ 0.8 \cdot f_{\text{correct}} - \frac{\max(0,\ |c| - 150)}{500}\right) \tag{4}$$

where $|c|$ is the character length of the completion and $f_{\text{correct}} = 1$ if $\hat{a} = a^*$, else 0.

## 4 Experimental Setup

To evaluate the impact of our reward shaping strategies, we configured our reinforcement learning environment and training pipeline as follows:

- **Base Model:** SmolLM2-360M, selected for its efficient parameter footprint, enabling rapid RL experimentation.
- **Dataset & Environment:** GSM8K (Grade School Math 8K), providing high-quality word problems to test reasoning trajectories.
- **Framework:** Hugging Face's Transformer Reinforcement Learning (TRL) library.
- **Optimization:** Low-Rank Adaptation (LoRA) via PEFT to minimize memory overhead and allow for feasible policy updates.
- **Hardware:** All training and evaluation phases were executed on a single Google Colab L4 GPU setup.

To evaluate each model, we sampled 600 examples from the GSM8K test set and measured accuracy (exact match on final answer), format adherence (presence of both <think> and <answer> tags), average total output length, and average <think> block length in tokens.

To properly isolate the impact of the GRPO interventions, we compared our RL-trained models against a zero-shot baseline (no prior task-specific training) and a Supervised Fine-Tuning (SFT) baseline.

## 5 Results

Figure 1 presents the quantitative metrics and token usage statistics across the baseline and our reward-shaped GRPO models, with detailed values reported in Table 1.

| Method | Accuracy | Format | Total Len. | Think Len. |
|---|---|---|---|---|
| SFT | **0.0417** | **0.4150** | 241.8 | 81.7 |
| GRPO-R1 | 0.0383 | 0.3967 | 243.5 | 76.3 |
| GRPO-R2 | **0.0417** | 0.3867 | 246.0 | 79.1 |
| GRPO-R3 | 0.0300 | 0.4033 | 243.9 | 78.4 |
| GRPO-R4 | 0.0250 | 0.3817 | 245.4 | **66.9** |

Table 1: Performance metrics and generation efficiency across SFT and GRPO reward variants, evaluated on 600 samples from the GSM8K test set.

**Quantitative Metrics:** The SFT baseline proved highly competitive, achieving the highest format adherence (0.4150) and tying for peak accuracy (4.17%) with GRPO-R2. Among the RL variants, GRPO-R1 (outcome-only) came closest to SFT accuracy (3.83%) but still fell short, consistent with

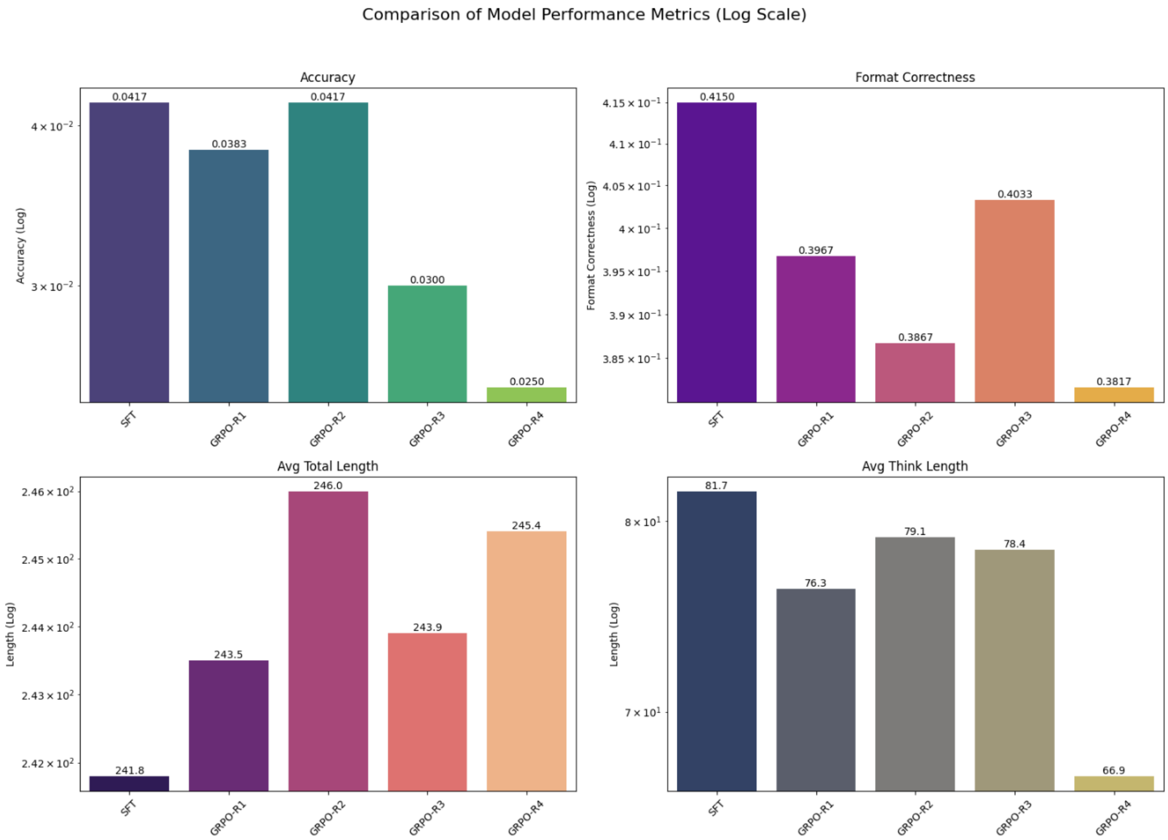


Figure 1: Comparison of accuracy, format adherence, average total length, and average think length across SFT and GRPO reward variants (R1–R4) in log scale. SFT and GRPO-R2 achieve the highest accuracy while R2 leads in format adherence. R4 produces the most concise reasoning chains at the cost of accuracy.

the hypothesis that sparse outcome rewards provide insufficient learning signal for small models over long generation horizons. GRPO-R2 matched SFT accuracy (4.17%) through its combined format and outcome reward, suggesting that denser reward signals help maintain the accuracy level established by SFT. However, GRPO-R3 and GRPO-R4 underperformed SFT at 3% and 2.5% respectively, indicating that at this model scale, more complex reward functions do not translate to accuracy gains and may in fact destabilize the policy.

**Efficiency and Token Usage:** The most consistent finding across all methods is R4's effect on generation length. R4 achieved the shortest average <think> length of all methods (66.9 tokens), representing an 18% reduction compared to SFT's 81.7 tokens — confirming that the length penalty successfully pressured the model toward more concise reasoning. Notably, this brevity came at the cost of accuracy (2.5%), reinforcing that for small models, forcing conciseness interferes with the reasoning process. R3 also underperformed despite its richer reward signal (3% accuracy, 78.4 think tokens), suggesting that rewarding intermediate arithmetic steps did not provide sufficient guidance for a 360M parameter model to generalize beyond what SFT already established.

## 6 Discussion

**SFT as a Robust Baseline:** The results across 600 evaluation samples reveal that SFT is a remarkably robust baseline that GRPO struggles to surpass at this model scale. SFT achieved the highest format adherence (0.4150) and tied for peak accuracy with only R2. This aligns with Chu et al. (2025), who note that SFT's direct imitation of structured examples produces strong formatting behavior — a property that RL with sparse or complex rewards tends to erode rather than improve.

**Impact of Reward Shaping:** The progression from R1 to R4 reveals a nuanced relationship between reward complexity and model performance at small scale. R1's near-SFT accuracy (3.83%) suggests that even minimal outcome signal preserves most of what SFT learned, while R2's tied accuracy (4.17%) shows that adding format guidance provides marginal benefit. R3 and R4's degraded accuracy suggests a capacity ceiling effect: the 360M parameter model lacks sufficient expressiveness to exploit complex, multi-component reward signals, and the additional reward terms introduce optimization noise that disrupts the policy rather than guiding it.

**The Cost of Explicit Length Penalization** R4 presents the clearest behavioral finding: the length penalty successfully reduced thinking length by 18% (81.7 to 66.9 tokens) but at the cost of the lowest accuracy across all methods (2.5%). This confirms that for small models, conciseness and correctness are in direct tension — the model cannot simultaneously shorten its reasoning and maintain answer quality. This contrasts with findings at larger scales where length penalties can improve efficiency without sacrificing accuracy.

**Why GRPO Did Not Surpass SFT** A key takeaway from this study is that GRPO's benefit is highly sensitive to model capacity. At 360M parameters, the model has a strict ceiling on mathematical generalization, and the reward signal — regardless of its design — cannot unlock reasoning strategies that the model is architecturally incapable of. The low base accuracy means most rollouts receive no positive reward, making the relative scoring in GRPO largely uninformative. This suggests that GRPO's advantages are best realized when the base model already has partial task capability to explore from.

## 7 Conclusion

This project investigated the impact of GRPO and reward shaping on the reasoning abilities of a 360M parameter language model evaluated across 600 GSM8K test samples. Our findings demonstrate that SFT remains a strong baseline that is difficult to surpass baseline at this model scale, achieving the highest format adherence and tying for peak accuracy with only the format-aware GRPO-R2 variant. No RL reward function decisively surpassed SFT on accuracy, suggesting that GRPO's benefits are gated by base model capacity — when the model rarely produces correct rollouts, the relative reward signal in GRPO becomes too sparse to guide meaningful policy improvement. The clearest behavioral effect of reward shaping was R4's 18% reduction in thinking length, though this came at the cost of accuracy, indicating that conciseness and correctness are in direct tension for small models.

**Future Work** Future work should scale this reward shaping framework to larger models (1B+ parameters) where partial task capability enables more informative reward signals and GRPO can explore meaningfully. Additionally, a hybrid reward combining R2's format signal with R3's step-aware signal may provide denser supervision than either alone, potentially stabilizing training at small scale. Finally, evaluating on the full GSM8K test set rather than a 600-sample subset would provide more statistically robust comparisons between reward variants.

## References

[1] Loubna Ben Allal, Anton Lozhkov, et al. 2025. SmolLM2: When Smol Goes Big – Data-Centric Training of a Small Language Model. *arXiv preprint arXiv:2502.02737*. https://arxiv.org/abs/2502.02737

[2] Tianshi Chu, Yuexiang Zhai, Jiantao Yang, Stephen Tong, Shaoan Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, and Yi Ma. 2025. SFT Memorizes, RL Generalizes: A Comparative Study of Foundation Model Post-training. *arXiv preprint arXiv:2501.17161*. https://arxiv.org/abs/2501.17161

[3] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training Verifiers to Solve Math Word Problems. *arXiv preprint arXiv:2110.14168*. https://arxiv.org/abs/2110.14168

[4] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxian Song, Mingchao Zhang, Y.K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. *arXiv preprint arXiv:2402.03300*. https://arxiv.org/abs/2402.03300

[5] Edward Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*. https://arxiv.org/abs/2106.09685